

Code-Layout, Readability and Reusability

Date	3 July 2026
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent indentation is maintained throughout the Python and Flask code.
2	Proper File Structure	Files and folders are logically organized	Yes	Project files are organized into folders such as dataset, models, templates, static, and application files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Meaningful variable names are used to improve readability and understanding.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named clearly based on their purpose, such as preprocessing, training, and prediction.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections of the code are documented using comments for better understanding.
6	Modular Design	Code is split into reusable functions/modules	Yes	Code is divided into reusable modules for preprocessing, model training, and prediction.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate and unnecessary code has been removed to improve maintainability.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling are implemented to manage runtime errors gracefully.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Model Training & Prediction	High
2	Feature Engineering Module	Python	Training Pipeline	High
3	Classification Model Module	Scikit-learn, XGBoost	Model Evaluation	High
4	Prediction Module	Flask, Pickle	Web Application	High
5	User Input Validation Module	Python, Flask	User Interface	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project follows a clean and modular folder structure.
Readability	5	Descriptive variable names and functions improve code readability.
Reusability	5	Functions and modules are reusable across training and deployment.
Documentation / Comments	5	Important code sections are well commented for easy understanding.
Overall Score	5	The project code is organized, maintainable, reusable, and follows good coding practices.